



Name _____

THE VIRTUAL EVOLUTION STICKLEBACK LAB

As you complete each part of the virtual lab, answer the questions below in the space provided.

OVERVIEW

Start by watching the video “Making of the Fittest: Evolution of the Stickleback Fish”. This is a really good example of the interplay between natural selection, molecular genetics and development in the evolution of a species.

<https://www.youtube.com/watch?v=Pv4Ca-f4W9Q>

1. At the end of the last glacial period, stickleback fish could be found in two habitats, open ocean and isolated lake. Fill in the table below with the major predator sticklebacks faced in each habitat and whether or not they had pelvic spines as a defense. (one point).

	Open ocean	Isolated lake
Major predator (name)		
Pelvic spines (Y/N)		

2. Describe the mutation that led to loss of pelvic spines? (one point, name the gene and the specific change)

VIRTUAL STICKLEBACK LAB

Then go to the interactive lab. You will need to complete Tutorials 1&2, Experiments 1&2 and Quizzes 1&2. You DO NOT need to complete the Analysis sections, Experiment 3 or the Final Quiz. The virtual lab will track your progress and record your quiz scores. As you go through each section, answer the questions below. Once you have finished the required parts, go to the progress tab and type your name in the Stickleback Evolution Lab Progress page and print to PDF. Attach that PDF to this assignment. Your name must appear in the PDF as proof you completed the assignment.

<https://media.hhmi.org/biointeractive/vlabs/stickleback2/index.html> then Click to enter the lab

INTRODUCTION

3. Read the Introduction and watch the videos to learn about why the stickleback was a good choice to study evolution. List two advantages of this model system. (one point)



TUTORIAL 1

Start Tutorial 1 so you can recognize a full pelvis, reduced pelvis and absent pelvis. The Ventral view is easiest. You don't have to be perfect, just learn the phenotypes before you move to experiment 1.

EXPERIMENT 1

As you go through each part of Experiment 1, background material will appear on the right in the Notebook. Use that to help you understand what you are doing. The lab will take you through the full process of treating and examining the fish, just as if you were doing this in the lab. In part 3, you can use the automated graphs. If it seems a little tedious, at least you didn't have to get your feet wet catching the fish!

When you've gone through parts 1, 2 & 3, you can take Quiz 1.

4. Which lake had more mutant (i.e., reduced or absent pelvis) fish? (one point)

5. How would you explain the result you observed in Q4? (one point)

TUTORIAL 2

Tutorial 2 will train you to recognize a full pelvis, reduced pelvis and absent pelvis in fossil specimens. This is a bit harder than looking at actual fish. You don't have to be perfect, just learn the phenotypes before you move to experiment 2.

EXPERIMENT 2

As you go through each part of Experiment 2, background material will appear on the right in the Notebook. Use that to help you understand what you are doing. The lab will look at fossils in different layers of rock (layer 2 is ~9000 years older than layer 5). In part 3, you can use the automated graphs. If it seems a little tedious, at least you didn't have to get your hands dusty splitting open the rock!

When you've gone through parts 1, 2 & 3, you can take Quiz 2.

6. What pattern do you see? (one point)

7. What can the fossil record tell us that studying live fish cannot? (one point)

After completing the quiz, click on the Progress tab.

Type your name in the box at upper right, then click on print.

Choose the "Save to PDF" option. Submit the PDF along with this assignment. (three points)